AI-Assisted Salinity Concentration Sensing via Diffraction Pattern Recognition of Optical Gratings

人工智能辅助的光栅衍射图样识别与盐度浓度检测

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1. Introduction

This project explores using artificial intelligence (AI) to recognize variations in grating diffraction patterns for monitoring environmental changes.

1.1 Grating

A diffraction grating, composed of parallel slits or grooves, produces interference fringes governed by the equation

$$d \sin\theta = \pm k\lambda$$

where d is the grating constant, θ is the diffraction angle, k is the diffraction order, and λ is the wavelength of the incident light. Constructive interference occurs at discrete angles, creating bright fringes. The spatial frequency, which grooves per unit length, determines angular dispersion: higher frequency means smaller d and wider fringe spread. Because high-frequency gratings respond sensitively to refractive-index perturbations, small environmental changes (*e.g.*, in salinity) shift fringe positions measurably. Thus, diffraction provides a sensitive optical means to detect refractive index variations.

1.2 Variable relationship

The grating was immersed in saline solutions of varying concentrations to simulate environmental conditions. Prior research (Zhao X. Y. et al., 2015) reports a near-linear relation between salinity M and refractive index n:

$$n = a + b\,M$$

As M rises, n increases which slows light propagation, alters optical path differences, and in turn shifts diffraction angles. Consequently, with greater salinity, fringes move toward the central axis and spacing

narrows, providing a quantitative optical indicator of concentration.

2. Results and Discussions

This study establishes a method to determine saline concentration by combining diffraction data with AI analysis (Zhao et al., 2015; Tengesdal et al., 2014). A robust dataset is critical for reliable prediction.

2.1 Dataset Acquisition

Finite-difference time-domain (FDTD) simulations (An et al., 2015) were first considered but proved computationally prohibitive. Instead, Python-based modeling generated 300 diffraction patterns within one day, simulating realistic optical setups: 3 μm line spacing, 5 μm groove width, and a 1 m propagation distance to a 2 × 2 m detection screen (Querry et al., 1977).

Random noise was added to mimic experimental uncertainty (Shafudah, 2015), preventing overfitting and enhancing real-world adaptability. Data were split 8:1:1 into training, validation, and test sets, ensuring effective learning and fair evaluation. This approach achieved computational efficiency while preserving experimental realism, bridging theoretical optics with practical AI modeling.

2.2 Data Preprocessing

To improve quality, the central diffraction region was cropped to remove irrelevant background. Wavelet-based denoising then decomposed each image via discrete wavelet transform; soft thresholding suppressed random noise yet preserved stripe features. The denoised reconstruction is shown in **Figure 1**.

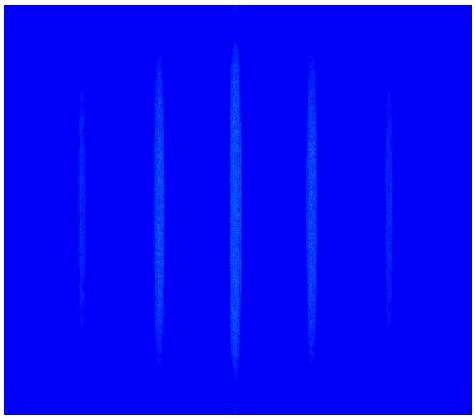


Figure 1 Difference Map Before and After Wavelet Filtering (Difference Amplified)

To emphasize periodic diffraction stripes, Sobel edge extraction was incorporated. Gradient magnitudes from horizontal and vertical filters produced an edge-enhanced map (**Figure 2**), which was normalized and concatenated with RGB channels to form a four-channel input. This fusion of intensity, color, and

structural cues allowed the regression model to detect subtle optical differences linked to salinity.

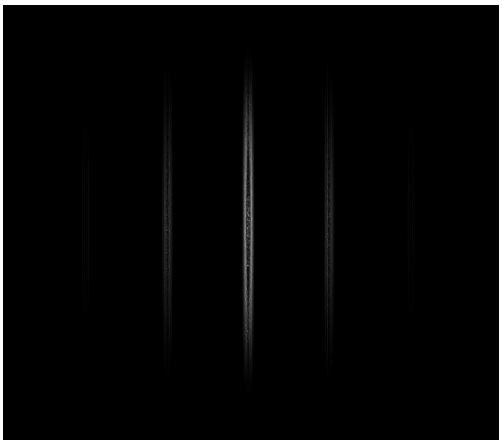


Figure 2 Result of Sobel Edge Extraction

2.3 Data Training

A regression model was trained to predict refractive index from images. Preprocessed inputs were fed to the model each epoch, and mean squared error (MSE) quantified prediction deviation. The Adam optimizer (learning rate 10^{-3}) updated parameters via backpropagation. Training and validation losses were logged per epoch, with checkpoints saving the lowest validation-loss model.

We used MobileNetV2 for its efficient depthwise–pointwise convolution design, which reduces parameters while maintaining accuracy and is ideal for optical applications. MSE served as the objective function, and PyTorch automatically handled gradient updates. Unlike classification settings, the network's final output dimension was modified to 1 to perform refractive-index regression. Pretrained weights were omitted due to domain mismatch between diffraction and natural-image data.

This training scheme integrated iterative optimization, loss monitoring, and checkpointing, ensuring stable convergence and reliable mapping between diffraction patterns and refractive indices.

2.4 Data Prediction

The 300-image dataset was divided into 240 training, 30 test, and 30 validation images. Over 50 epochs, average test loss decreased steadily (log scale not shown). The trained model's refractive-index predictions closely matched true values, with all errors within 2% and an average test MSE = 0.0038. The prediction loss curve is plotted in **Figure 3**.

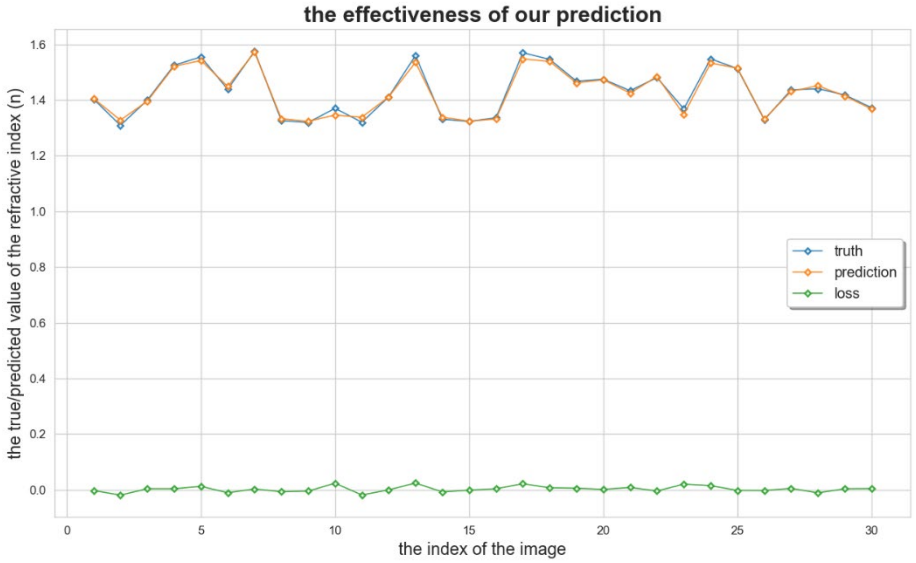


Figure 3 The Effectiveness of Our Prediction

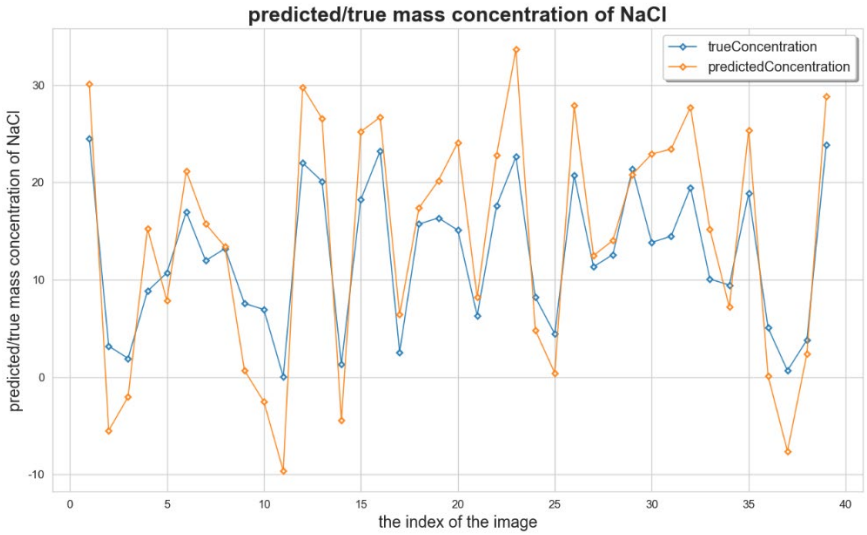


Figure 4 Comparison of the Salt Concentration Predicted by the Model with True Values

Using the empirical relation $n = 1.33 + 0.0016M$, predicted refractive indices were converted into corresponding salt concentrations. **Figure 4** compares predicted and true values, revealing strong agreement and confirming the model's capacity to extract concentration-related optical cues. Minor deviations suggest a room for further optimization in data augmentation and model tuning.

3. Conclusion

This study demonstrates a feasible framework for AI-assisted optical salinity measurement. By modeling diffraction physics and using efficient CNN architectures, it accurately associates diffraction-pattern variations with refractive-index shifts. Compared with traditional computational methods, the Python-based simulation approach drastically reduced data-generation time while retaining physical realism. Wavelet and edge preprocessing improved pattern clarity and robustness, and MobileNetV2 efficiently captured subtle features. The resulting model achieved high prediction accuracy with minimal error, confirming the potential of lightweight AI models for real-time environmental monitoring through optical sensing. Future work may extend to broader refractive-index ranges, multi-wavelength gratings, or hybrid AI–physics models for enhanced interpretability.

References

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